

Graphing Quadratic Expressions (Part IV) Homework

Circle the irrational number in each list below.
There is only one irrational number in each list.
You may use a calculator to convert to decimal form if helpful.

① $\sqrt{36}$, -5.567 , $\frac{4\pi}{2\pi}$, $4\frac{6}{7}$, -5π

② $\frac{\sqrt{2}}{5}$, $\frac{2}{3}$, $(\sqrt{6})^2$, $1.0\overline{8}$, $\sin 150^\circ$

③ $\sqrt{\frac{4}{9}}$, 0.0009898 , $\tan 47^\circ$, $\cos 90^\circ$, $\sqrt{25}$

④ $\sin 180^\circ$, $\sqrt{\pi}$, $\frac{\sqrt{3}}{2\sqrt{3}}$, $\sin 270^\circ$, $\frac{13}{59}$

Graph each quadratic expression. Show the coordinates of the vertex, y-intercept, and x-intercepts (if there are any). Draw the line of symmetry (as a dotted line) and write its equation.

⑤ $y = (x+3)^2 - 5$

⑥ $y = -2x^2 + 12$

⑦ $2(x+1)^2 - 10 = y$

⑧ $4(x+2)^2 - 5 = y$

⑨ $-4(x-1)^2 + 2 = y$

⑩ $y = 3(x-1)^2 - 5$

⑪ $y = x^2 - 10x + 16$

⑫ $x^2 + 2x - 3 = y$

⑬ $y = x^2 - 4x + 3$

⑭ $x^2 - 2x - 15 = y$

⑮ $y = x^2 - 6x + 8$

⑯ $x^2 - 8x + 7 = y$

⑰ $y = -x^2 - 8x - 12$

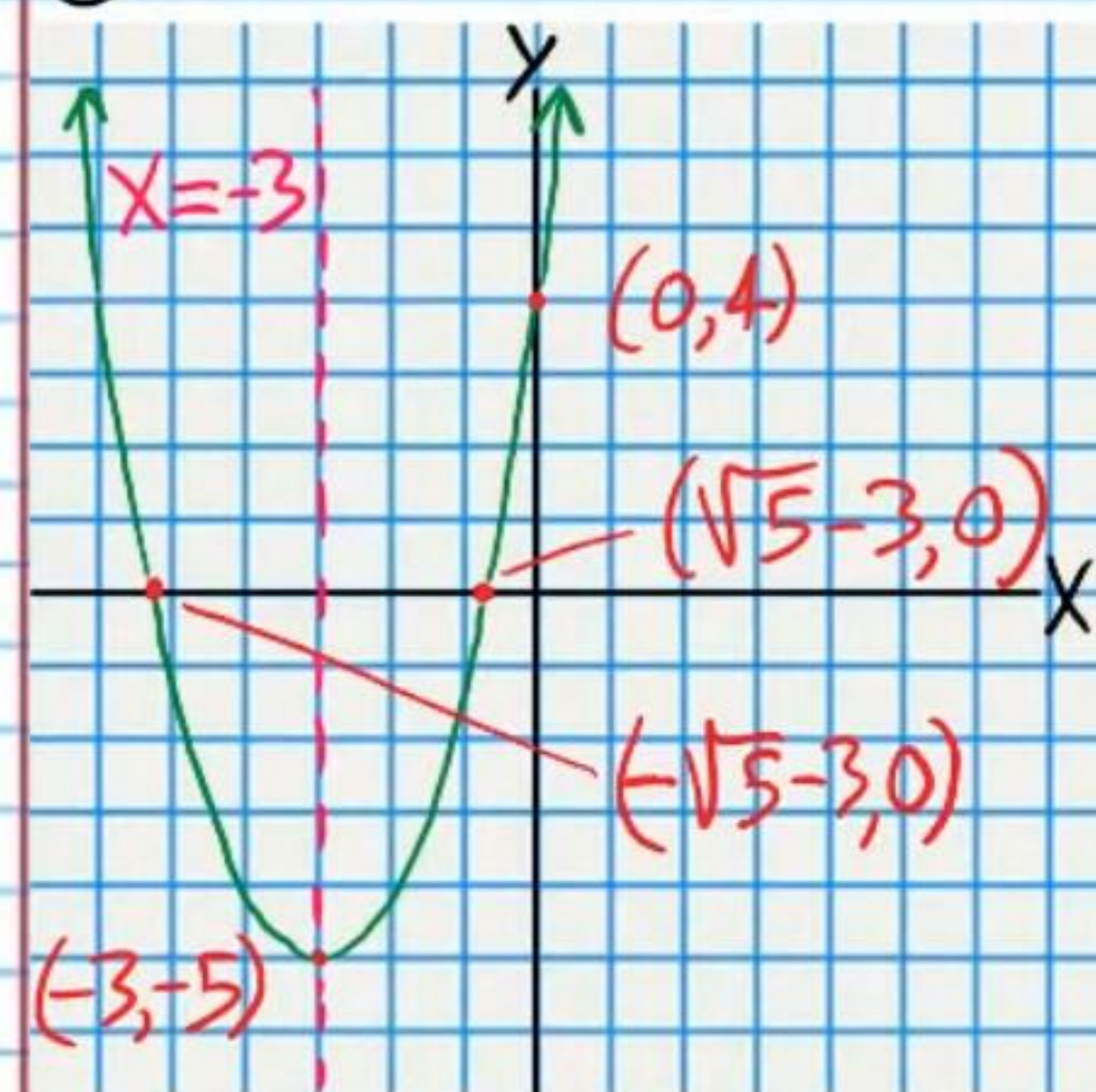
⑱ $-x^2 - 2x + 8 = y$

⑲ $y = -x^2 - 4x + 5$

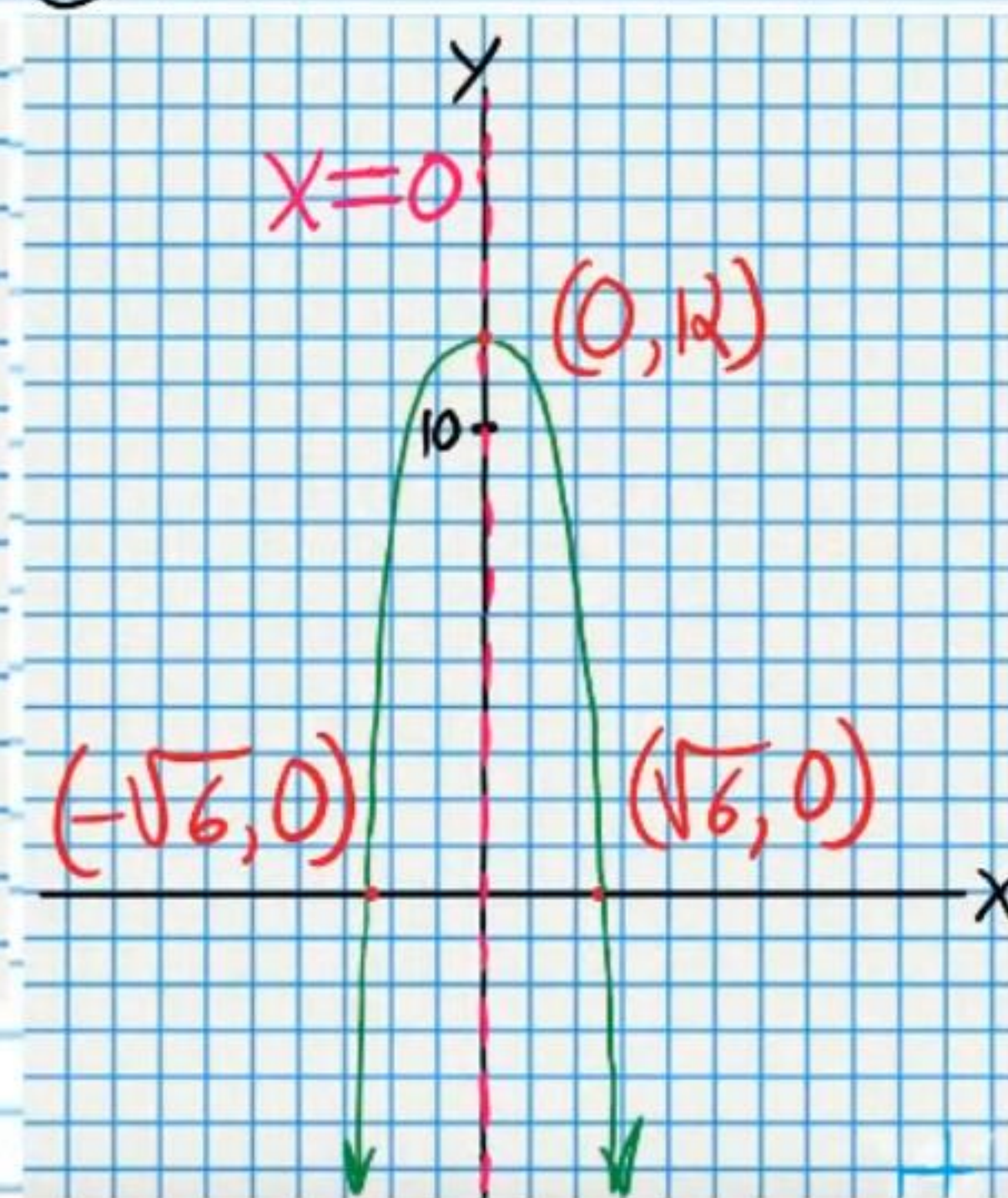
Answers

- ① -5π ② $\frac{\sqrt{2}}{5}$ ③ $\tan 47^\circ$ ④ $\sqrt{\pi}$

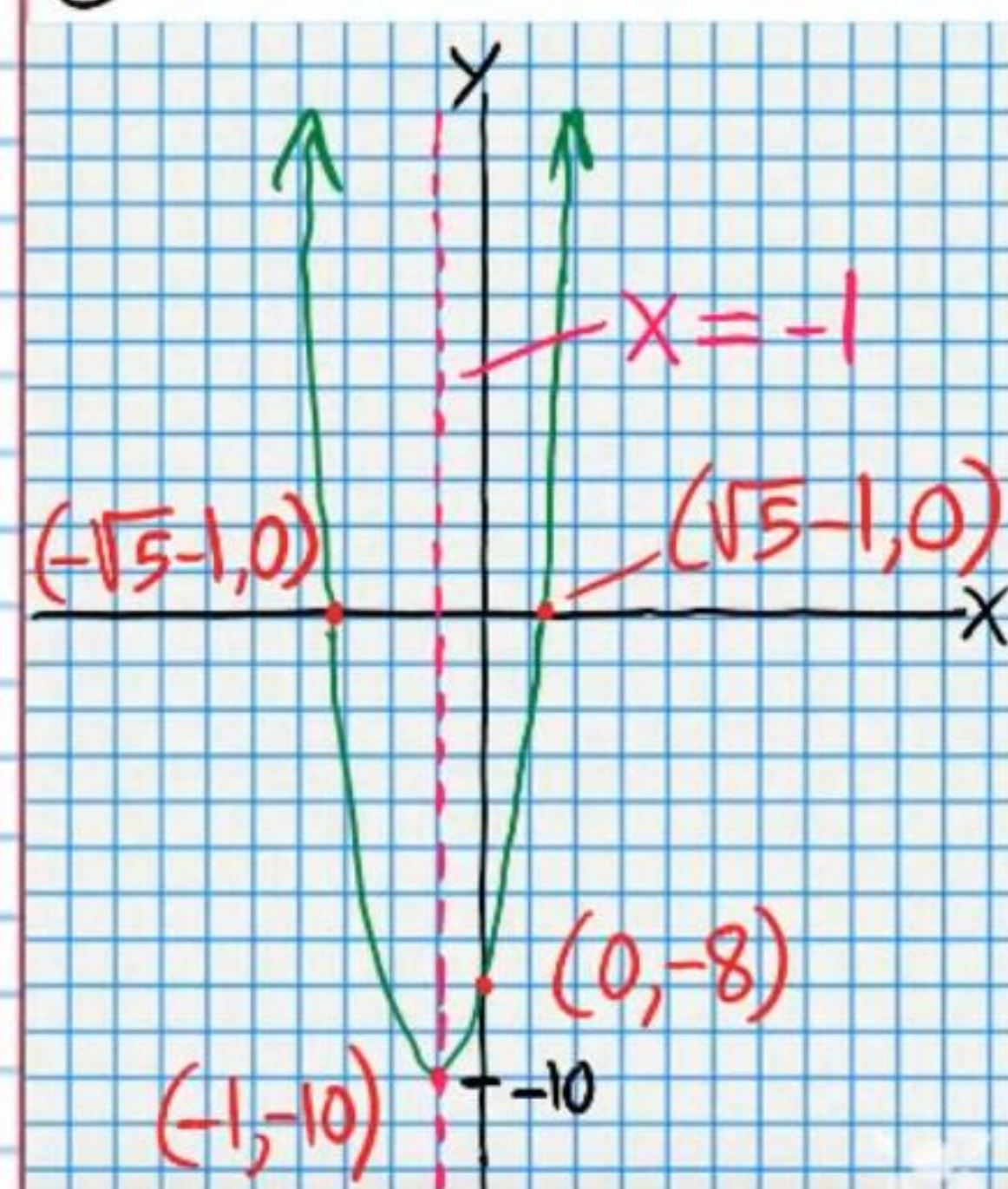
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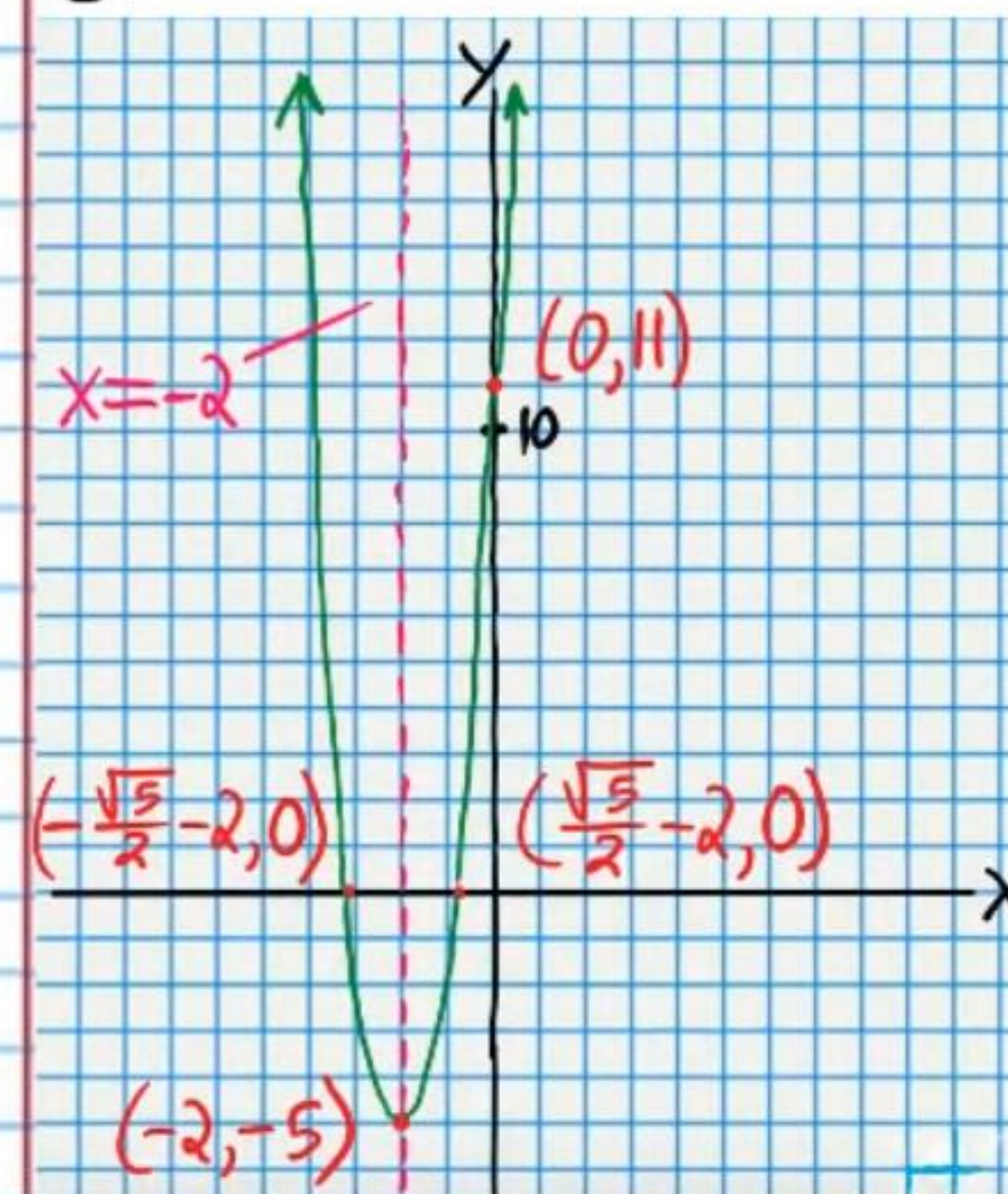
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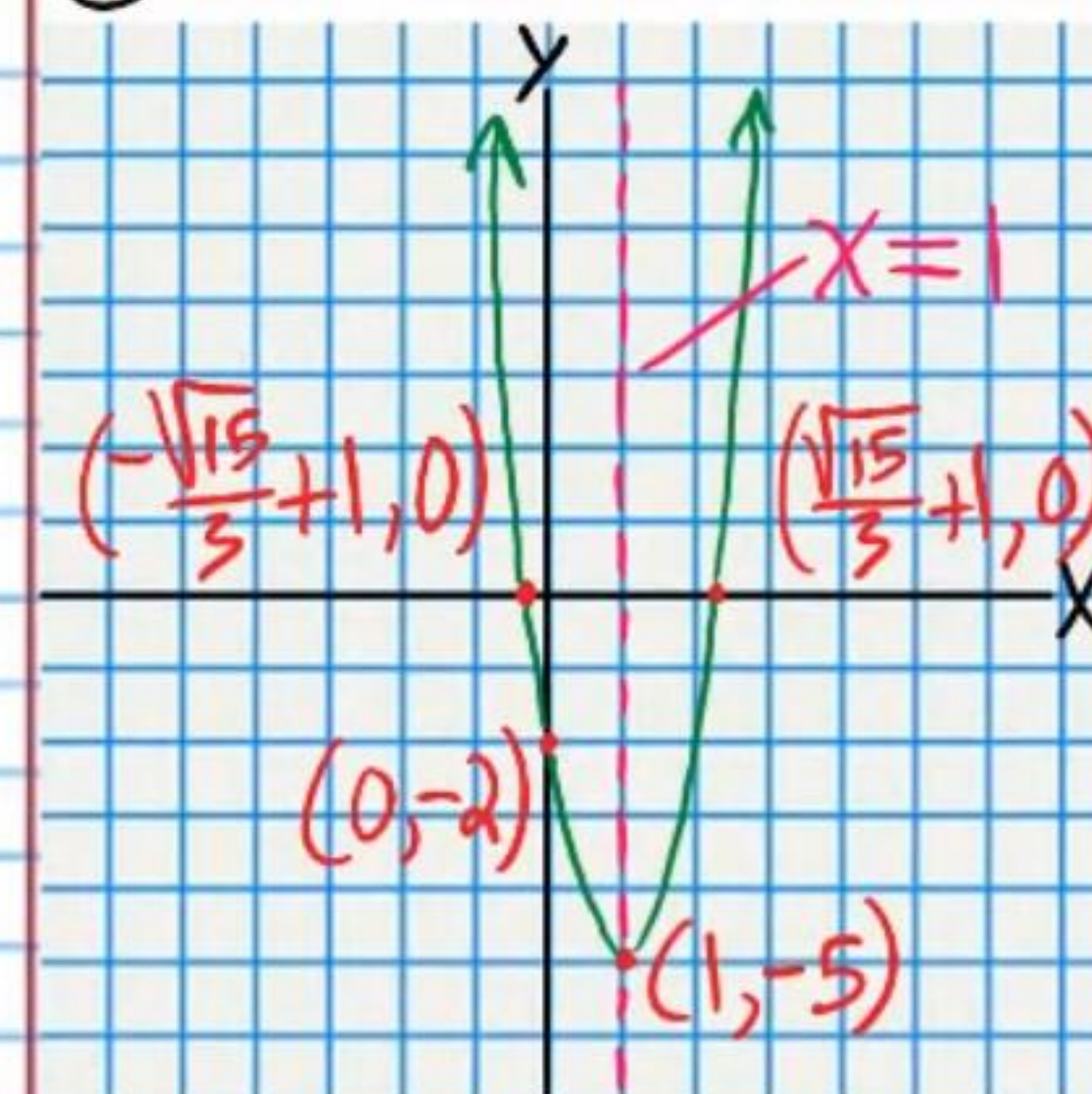
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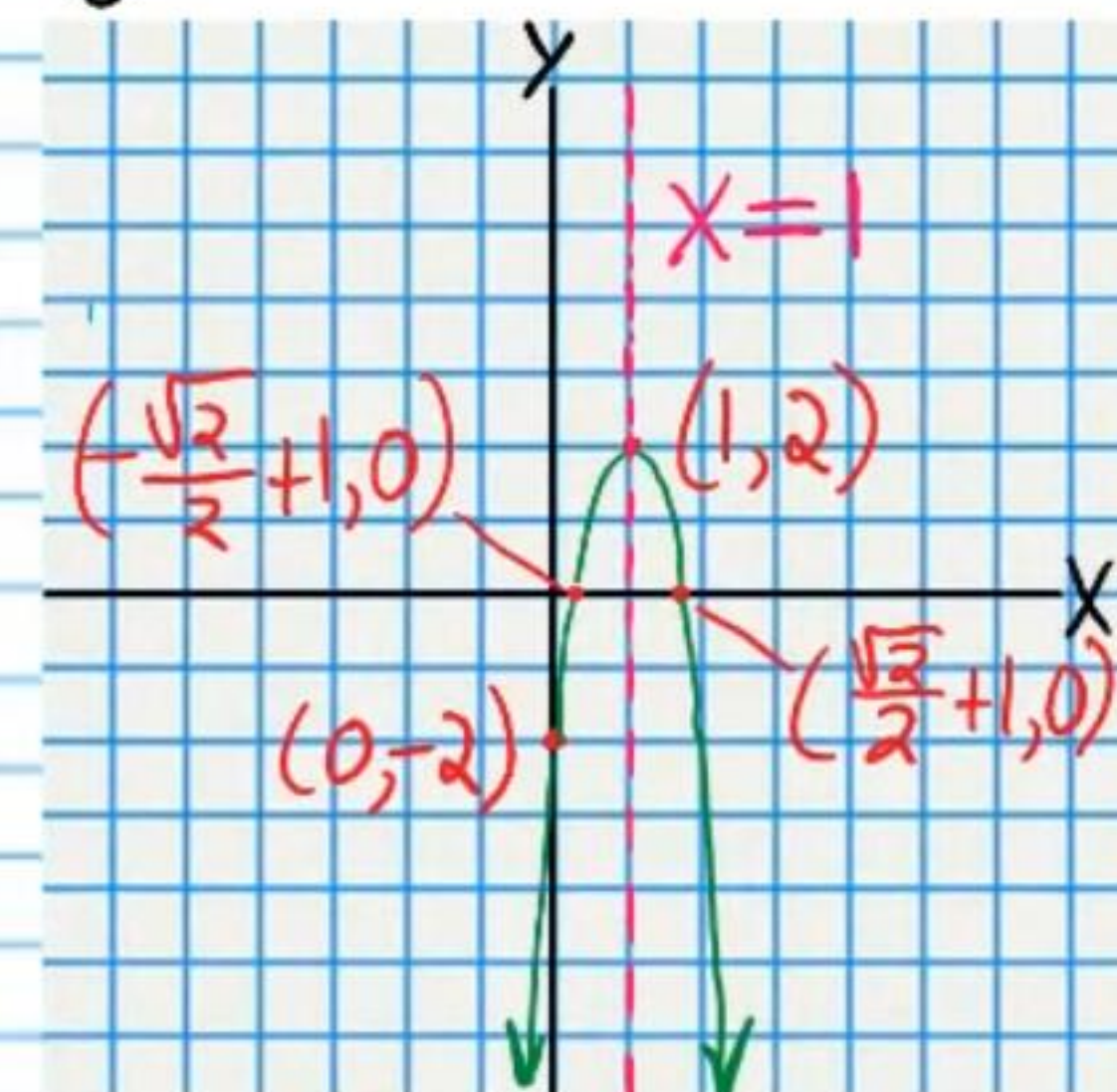
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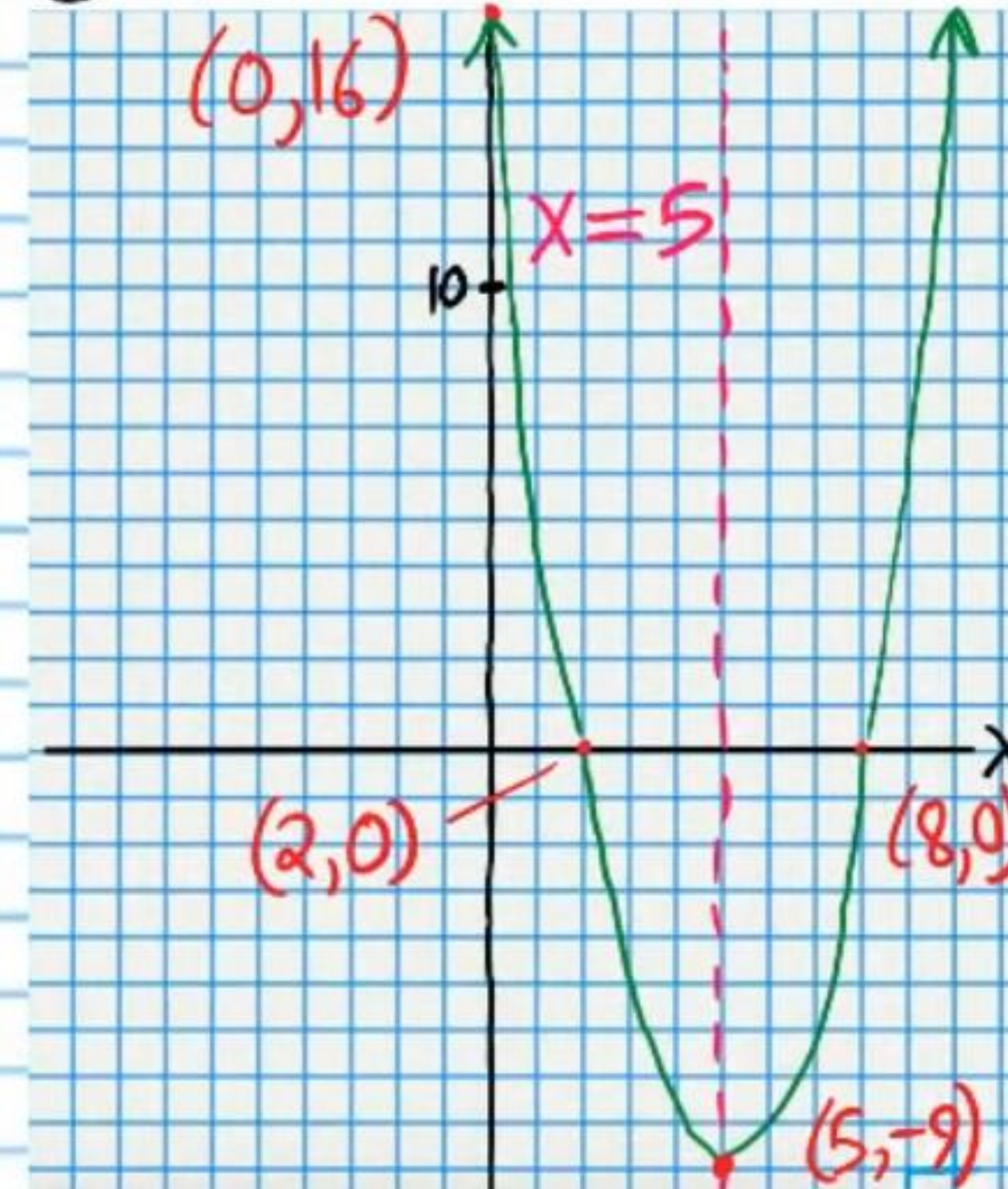
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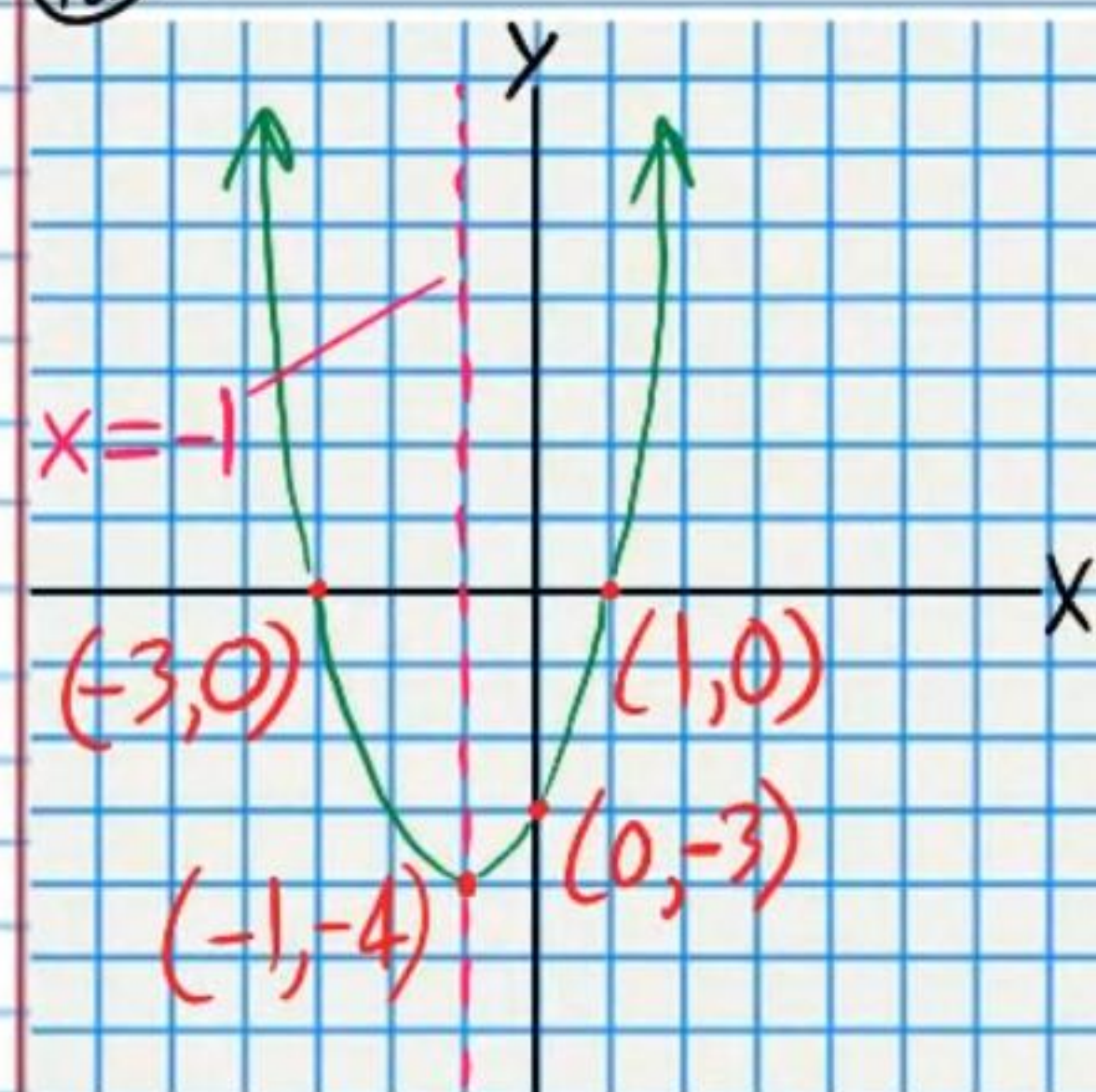
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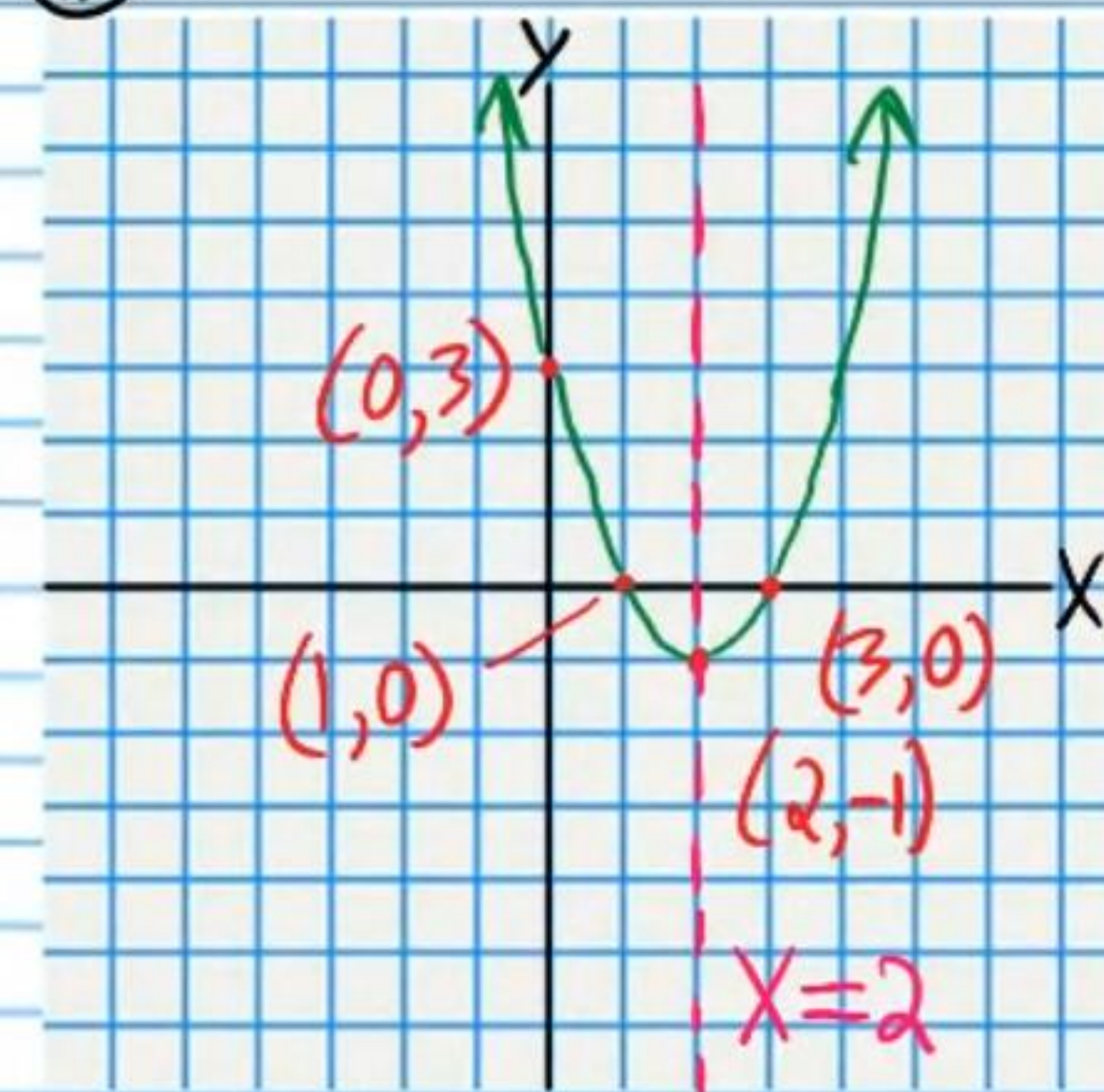
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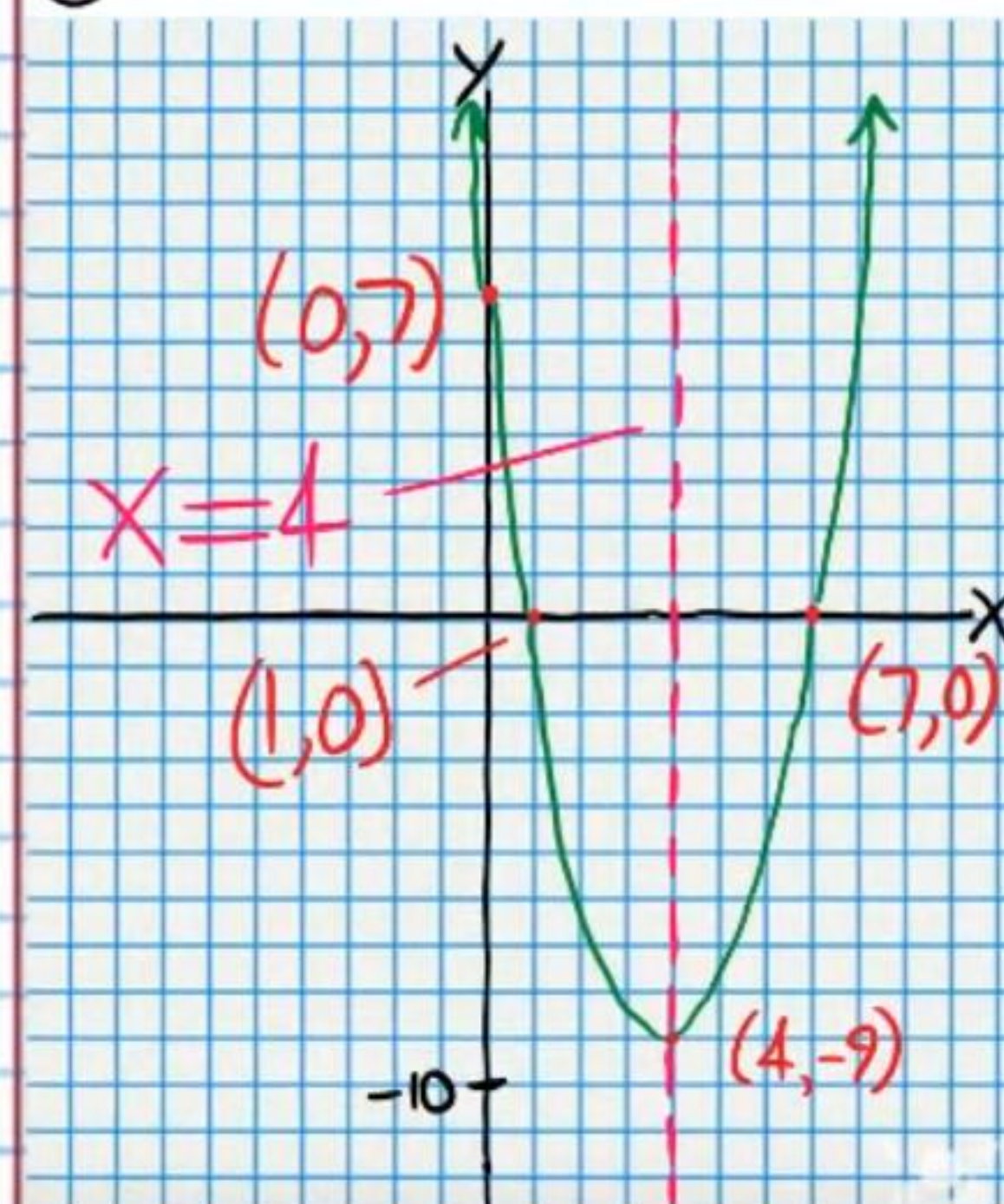
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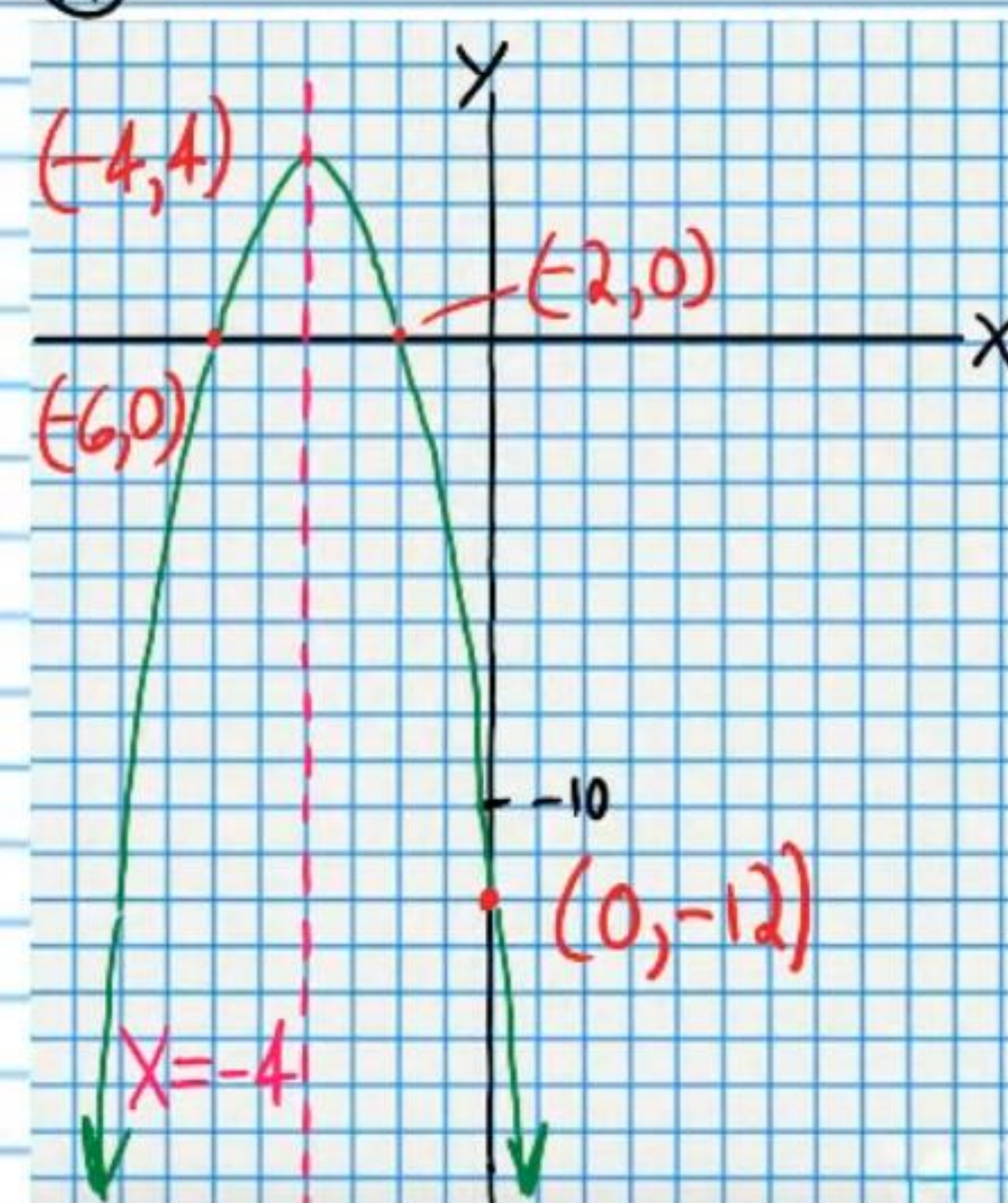
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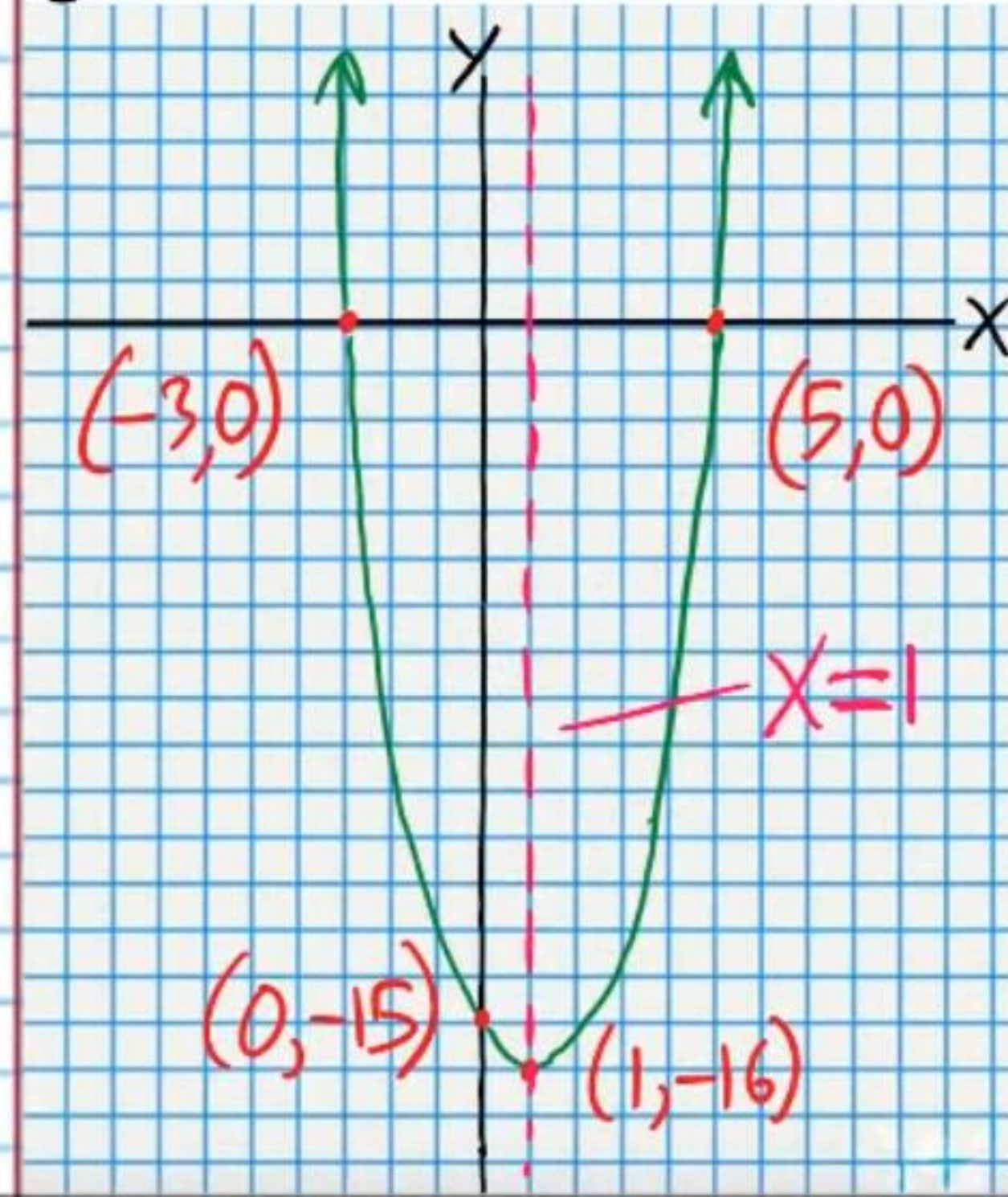
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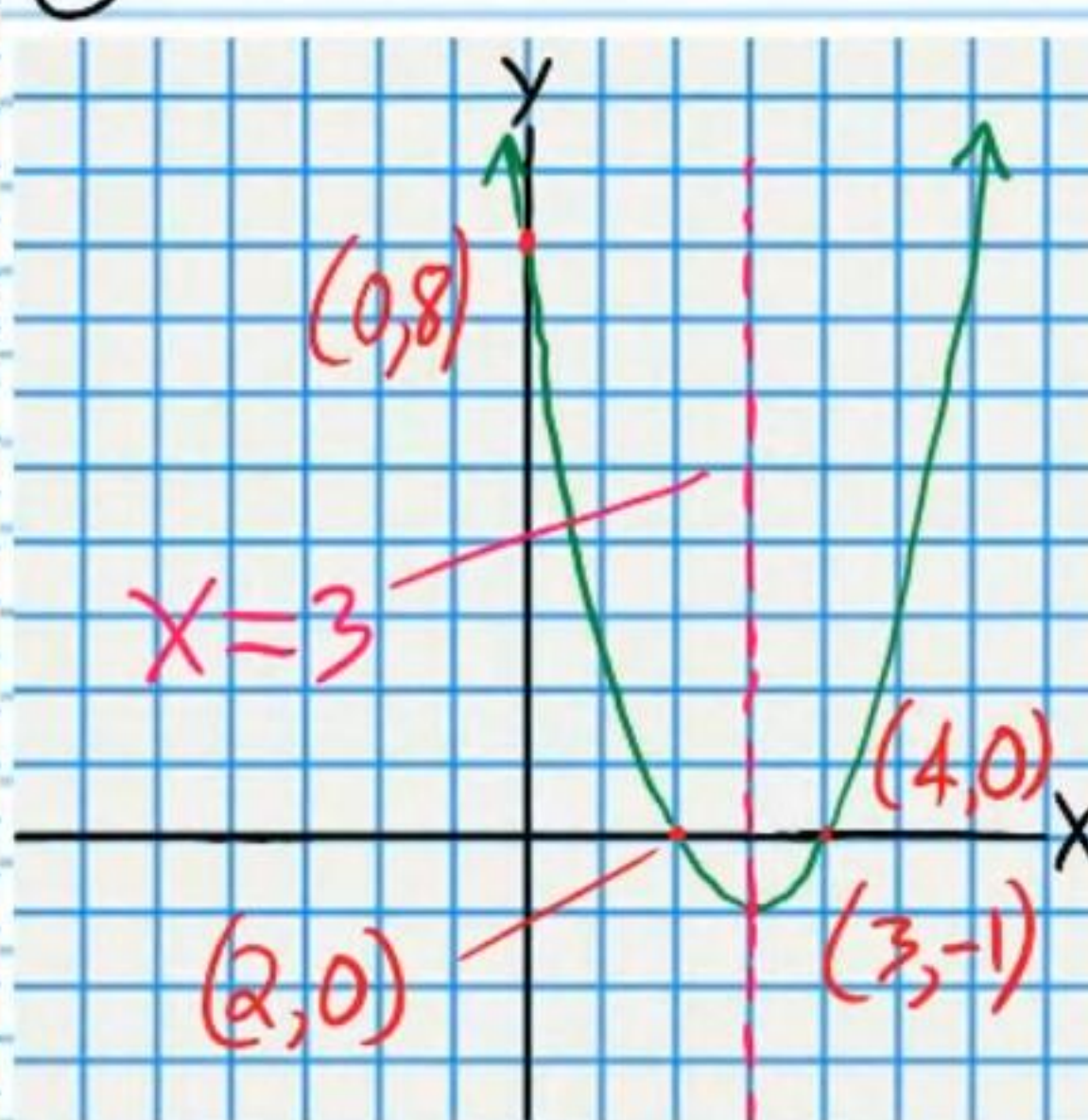
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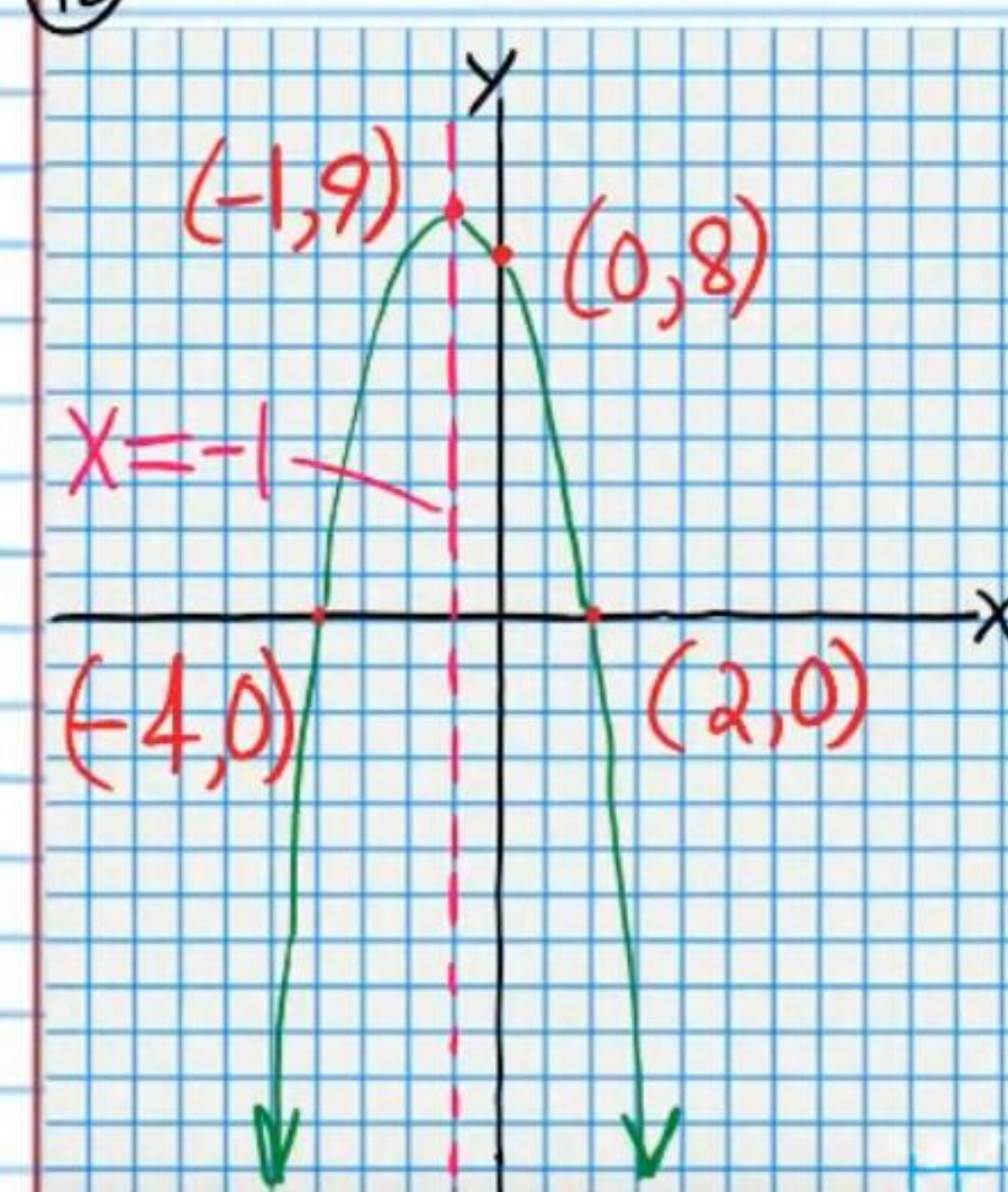
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